

New values of the largest Littlewood–Richardson coefficient via iterated Okounkov compression

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September 3, 2026

Abstract

Let $C(n) = \max c_{\mu\nu}^\lambda$ over all triples with $|\mu| + |\nu| = |\lambda| = n$ (OEIS A387851). Pak and Soskin recently computed $C(n)$ for $24 \leq n \leq 45$ by searching all partition pairs at every split k , $\sum_{k \leq n/2} p(k)p(n-k)$ pairs in all; the last degree took 20 hours on 32 cluster cores. They conjectured that every maximum sits on a nested pair $\mu \subseteq \nu \subseteq \lambda$ with ν/μ a disjoint union of squares, uniquely so up to symmetry. Our starting point is the Okounkov inequality, proved by Lam, Postnikov and Pylyavskyy, which bounds $c_{\mu\nu}^\lambda$ by the coefficient of the pair obtained by halving the sum vector $w = \mu + \nu$. Together with conjugation symmetry it gives a map T on partitions of n that never decreases the largest coefficient available at a sum vector. Pak and Soskin’s structure theorem applies the Lam–Postnikov–Pylyavskyy inequality once, then the averaging step twice with a conjugation in between; we simply keep iterating until the orbit closes, and that turns the inequality into a search space reduction. The $p(n)$ sum vectors collapse onto a much smaller set of terminal vectors, expanding the terminal pairs alone already determines $C(n)$, and a far smaller scan over the preimages of the winning terminal vectors provably recovers every maximizing triple. The resulting algorithm reproduces all 22 reported values $C(24), \dots, C(45)$ on a single laptop core, taking 34 minutes end to end at $n = 45$, and extends the sequence by eight terms,

$$C(46), \dots, C(53) = 5932, 8481, 9926, 12828, 15762, 18570, 22982, 31264.$$

The accompanying maximizer search is exhaustive rather than heuristic, and both parts of the Pak–Soskin conjecture are verified at each new value: every maximal triple is square-union nested, and the maximizing triple is unique up to interchange of μ and ν and simultaneous conjugation.

2020 Mathematics Subject Classification.

Primary 05E05; Secondary 05A17, 05-08, 20C30.

Keywords. Largest Littlewood–Richardson coefficient, Okounkov inequality, Schur positivity, integer partitions, computational combinatorics, OEIS A387851.

1 Introduction

For partitions μ, ν, λ with $|\mu| + |\nu| = |\lambda|$, the Littlewood–Richardson coefficient $c_{\mu\nu}^\lambda$ is the multiplicity of the Schur function s_λ in $s_\mu s_\nu$. Following a question of Stanley [11, Exc. 82(c)],

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set

$$C(n) = \max\{c_{\mu\nu}^\lambda : |\mu| + |\nu| = |\lambda| = n\},$$

the largest Littlewood–Richardson coefficient at degree n ; the sequence is OEIS A387851 [6]. Stanley showed that the maximum is attained at $|\mu| \sim n/2$ and that $C(n) = 2^{n/2 - O(\sqrt{n})}$, and asked for the asymptotic limit shape; Pak, Panova and Yeliussizov [8] answered that question, proving that all three partitions have Vershik–Kerov–Logan–Shepp shape, and tabulated $C(n)$ for $n \leq 23$. Pak and Soskin [9] computed the full table of $C(n, k) = \max\{c_{\mu\nu}^\lambda : |\mu| = k, |\lambda| = n\}$ for $24 \leq n \leq 45$ by direct search over triples of partitions, reported that $n = 45$ required 20 hours on 32 cores, and stated that going beyond appeared infeasible by their method. Based on the data they conjectured:

Conjecture 1.1 (Pak–Soskin [9, Conj. 1.1]). *For $n \geq 16$, every triple attaining $C(n)$ satisfies $\mu \subseteq \nu \subseteq \lambda$ (up to swapping μ, ν) with ν/μ a disjoint union of square shapes; for $n \geq 28$ the maximal triple is unique up to interchange of μ and ν and simultaneous conjugation.*

Here and throughout, *conjugation* conjugates all three partitions simultaneously, and a triple is *square-union nested* if, after possibly interchanging μ and ν , it satisfies $\mu \subseteq \nu \subseteq \lambda$ and every edge-connected component of ν/μ is a full $c \times c$ square.

This note extends the computation by eight terms and verifies Conjecture 1.1 at each of them.

Theorem 1.2 (computer-assisted). $C(46), \dots, C(53) = 5932, 8481, 9926, 12828, 15762, 18570, 22982, 31264$. *At each of these n , every maximal triple is square-union nested and the maximizing triple is unique up to interchange of μ and ν and simultaneous conjugation, so both parts of Conjecture 1.1 hold for $46 \leq n \leq 53$.*

Section 2 proves that the search underlying Theorem 1.2 is exhaustive, and Section 3 reports the gates the implementation was put through, including an exhaustive brute force comparison at every degree $2 \leq n \leq 28$.

The engine is an inequality conjectured by Okounkov [7] and proved in full generality by Lam, Postnikov and Pylyavskyy [3, Thm. 11]. The inequality is theirs; what is done here is to iterate it. In the proof of their Theorem 1.2, [9] first apply the Lam–Postnikov–Pylyavskyy inequality once and then the Okounkov averaging step twice, with a conjugation in between, to exhibit *some* maximizer of a prescribed shape. Applying the averaging step repeatedly, until the induced map on sum vectors closes up on itself, turns the inequality into a compression map that shrinks the search space by more than three orders of magnitude and, more importantly, leaves a provably complete list of candidates for the maximizers. That is the entire content of the method, described next.

The Schur positivity landscape around this inequality has moved quickly in 2026. Chan, Chen, Pak and Soskin [1] generalized the Ahlswede–Daykin inequality to a Schur positive ADS inequality that contains the Lam–Postnikov–Pylyavskyy inequality as a special case. Speyer [10] proved the log-concavity conjecture of Lam, Postnikov and Pylyavskyy, Le and Nguyen [4] have since extended that result to skew Schur functions, and, closest to home, Chan and Pak [2] determined equality conditions for correlation inequalities, the Okounkov inequality among them. None of that machinery is needed below: the argument uses the coefficientwise inequality (O) and nothing else. What is new here is the decision to iterate the averaging and conjugation step on sum vectors, and to treat the finite dynamics that fall out as an exact computational compression.

2 The compression map and why the search is complete

Partitions are padded with trailing zeros where needed, and sums such as $\mu + \nu$ are entrywise. Write $\lfloor w/2 \rfloor$ and $\lceil w/2 \rceil$ for the entrywise floor and ceiling halves of an integer vector w , and conj for partition conjugation. Two classical facts are used, and nothing else:

(O) (Okounkov inequality [7, 3]) For all partitions μ, ν and every λ ,

$$c_{\mu\nu}^{\lambda} \leq c_{\lfloor (\mu+\nu)/2 \rfloor, \lceil (\mu+\nu)/2 \rceil}^{\lambda}.$$

(C) (Conjugation symmetry) $c_{\mu\nu}^{\lambda} = c_{\text{conj } \mu, \text{conj } \nu}^{\text{conj } \lambda}$.

The right-hand side of (O) depends on (μ, ν) only through the sum vector $w = \mu + \nu$, which is itself a partition of n . For a partition w of n put

$$M(w) = \max_{\lambda \vdash n} c_{\lfloor w/2 \rfloor, \lceil w/2 \rceil}^{\lambda}, \quad T(w) = \text{conj}(\lfloor w/2 \rfloor) + \text{conj}(\lceil w/2 \rceil).$$

Thus T halves w , conjugates both halves, and records the new sum vector; $T(w)$ is again a partition of n .

Lemma 2.1. *Let w be a partition of n . Then*

- (a) $M(w) = \max\{c_{\mu\nu}^{\lambda} : \mu + \nu = w\}$, the maximum being over all pairs of partitions with sum vector w and all λ ;
- (b) $M(w) \leq M(T(w))$.

Proof. (a) The pair $(\lfloor w/2 \rfloor, \lceil w/2 \rceil)$ has sum vector w , so $M(w)$ is at most the right-hand side. Conversely, if $\mu + \nu = w$ then (O) gives $c_{\mu\nu}^{\lambda} \leq c_{\lfloor w/2 \rfloor, \lceil w/2 \rceil}^{\lambda} \leq M(w)$ for every λ .

(b) Choose λ with $c_{\lfloor w/2 \rfloor, \lceil w/2 \rceil}^{\lambda} = M(w)$. By (C),

$$M(w) = c_{\text{conj } \lfloor w/2 \rfloor, \text{conj } \lceil w/2 \rceil}^{\text{conj } \lambda}.$$

The pair $(\text{conj } \lfloor w/2 \rfloor, \text{conj } \lceil w/2 \rceil)$ is a pair of partitions with sum vector $T(w)$, by the definition of T . Applying part (a) at $T(w)$ bounds the right-hand side by $M(T(w))$. $\square$

Since there are only $p(n)$ partitions of n , the sequence $w, T(w), T^2(w), \dots$ must repeat. Let $k(w) \geq 1$ be least with $T^{k(w)}(w) \in \{w, T(w), \dots, T^{k(w)-1}(w)\}$ and put

$$w^* = T^{k(w)}(w),$$

the first repeated iterate. By construction w^* recurs under T , so it is a periodic point of T ; we call it the *terminal vector* of w , and call $(\lfloor w^*/2 \rfloor, \lceil w^*/2 \rceil)$ the *terminal pair*. Iterating Lemma 2.1(b) along $w, T(w), \dots, T^{k(w)}(w)$ gives

$$M(w) \leq M(w^*) \quad \text{for every } w \vdash n. \tag{1}$$

Proposition 2.2. *Let $n \geq 1$. Then*

- (a) $C(n) = \max_{w \vdash n} M(w^*)$, the maximum being over the terminal vectors alone;

(b) if $c_{\mu\nu}^\lambda = C(n)$ and $w = \mu + \nu$, then $M(w^*) = C(n)$.

Proof. By Lemma 2.1(a), $C(n) = \max_{w \vdash n} M(w)$.

(a) Each w^* is itself a partition of n , so $\max_w M(w^*) \leq \max_w M(w) = C(n)$; and (1) gives $M(w) \leq M(w^*) \leq \max_w M(w^*)$ for every w , hence $C(n) \leq \max_w M(w^*)$.

(b) Lemma 2.1(a) gives $M(w) \geq c_{\mu\nu}^\lambda = C(n)$, and (1) gives $M(w^*) \geq M(w) \geq C(n)$, while (a) gives $M(w^*) \leq C(n)$. $\square$

Call a terminal vector *winning* if $M(w^*) = C(n)$, and call a partition w of n a *suspect* if w^* is winning.

Corollary 2.3. *Every triple (λ, μ, ν) with $c_{\mu\nu}^\lambda = C(n)$ has $\mu + \nu$ a suspect. Consequently, expanding $s_\mu s_\nu$ for every decomposition $\mu + \nu = w$ of every suspect w recovers every maximal triple, and the resulting verification of Conjecture 1.1 is exhaustive, not heuristic.*

Proof. Immediate from Proposition 2.2(b) and the definitions. $\square$

Remark 2.4 (the iteration does not converge to a fixed point). Nothing above asserts that w^* is fixed by T , and in general it is not. At six of the eight degrees treated below the sum vector of the maximizing pair lies on a genuine 2-cycle of T , and only at $n = 46$ and $n = 53$ is it a fixed point. Only Lemma 2.1(b) and the finiteness of the state space are used, so the argument is indifferent to which. Two directly verified facts explain two of the numbers reported below. First, at every degree $46 \leq n \leq 53$ the set of terminal vectors was verified to be exactly the set of partitions of n whose odd parts are pairwise distinct, so the terminal class counts in Table 1 are values of OEIS A006950 [6]. Second, see Lemma 2.5. The dynamics of T in their own right are taken up in a separate note; nothing in the present one depends on them.

Lemma 2.5. *M is constant on every cycle of T . Hence the set of winning terminal vectors is a union of T -cycles, and its cardinality is a sum of cycle lengths.*

Proof. If $T^m(v) = v$ then Lemma 2.1(b) gives $M(v) \leq M(T(v)) \leq \dots \leq M(T^m(v)) = M(v)$, so all the inequalities are equalities. $\square$

At each degree $46 \leq n \leq 53$ the winning set turns out to be a *single* cycle: a 2-cycle except at $n = 46$ and $n = 53$, where it is a fixed point. This is the column “classes attaining” of Table 2, whose entries 1 and 2 are therefore an artefact of the cycle structure and not a failure of uniqueness. At the six degrees with a 2-cycle, the two winning terminal pairs are conjugates of one another, so they carry the same maximal triples up to conjugation.

The computation

1. **Compress.** For each $w \vdash n$ compute w^* by iterating T (memoized); record the fibers $w \mapsto w^*$. Cost: seconds.
2. **Expand terminal pairs.** For each distinct terminal vector w^* , expand $s_{\lfloor w^*/2 \rfloor} s_{\lceil w^*/2 \rceil}$ with `lrcalc` [5] and take the largest coefficient; this is $M(w^*)$. By Proposition 2.2(a) the maximum over terminal vectors is $C(n)$. This pass dominates the runtime.
3. **Exhaust the suspects.** Expand $s_\mu s_\nu$ for every decomposition $\mu + \nu = w$ of every suspect w . By Corollary 2.3 this recovers every maximal triple (λ, μ, ν) .

The $p(n)$ sum vectors collapse onto a much smaller set of terminal vectors: 12 306 at $n = 46$ growing to 32 490 at $n = 53$, against $2.1 \cdot 10^7$ pairs (μ, ν) with $|\mu| \leq |\nu|$ at $n = 46$ growing to $1.0 \cdot 10^8$ at $n = 53$. In pass 3 the suspects are few: tens of sum vectors and 10^3 to 10^4 pairs at every degree computed.

Each maximal triple is then tested for the square-union property (each edge-connected component of the skew shape ν/μ is a full $c \times c$ square, the reading of [9] matching their $n = 40$ example, in which four unit boxes touching only diagonally count as four squares), and the list is collapsed under the interchange and conjugation action of Conjecture 1.1 to count orbits.

3 Validation

The method was gated five ways before any new value was believed.

(V1) Exhaustive brute force at every degree $2 \leq n \leq 28$. For each such n we computed $C(n)$ and the complete list of maximal triples by direct search over every pair $(\mu \vdash k, \nu \vdash n - k)$ with $k \leq n/2$, sharing no step with the compression, and compared both against the three-pass method. At all 27 degrees the values agree and the two lists of maximal triples agree *as sets*; so does the verdict on each part of Conjecture 1.1. This is a direct test of Corollary 2.3: had the suspect scan been able to miss a maximizer, the lists would differ. Script `bruteforce_gate.py`, log `bruteforce_gate.log`.

(V2) The full reported range. The implementation reproduces all 22 values $C(24), \dots, C(45)$ of the appendix of [9], in one run. Those values were not transcribed: they were parsed out of the L^AT_EX source of the arXiv e-print of [9], and the maxima of the parsed $C(n, k)$ grid were checked against the parsed $C(n)$ row (all 22 rows, no discrepancy). Runtimes at the degrees [9] timed:

n	$C(n)$ [9]	$C(n)$ here	runtime [9]	runtime here (1 core)
25	45	45	28 s (4 cores)	0.8 s
30	176	176	198 s (4 cores)	4.6 s
35	423	423	3 258 s (4 cores)	40 s
40	1 484	1 484	44 919 s (4 cores)	234 s
41	1 768	1 768	5 609 s (32 cores)	387 s
45	4 323	4 323	72 172 s (32 cores)	2 042 s

The $n = 41$ row is the one where [9] moved from a laptop to the cluster, which is why their time drops there. All six of our times come from the single run recorded in `revalidate_24.45.log`, and all six are totals over the three passes. At $n = 45$ the time splits as 15 s of compression, 1 429 s of terminal expansion and 598 s of suspect scanning. Our times are single core, and the machine was carrying heavy unrelated load throughout, so they are upper bounds on the cost of the method rather than benchmarks. The same caveat applies to Table 1, whose degrees were run at different times and under different load, which is why $n = 51$ there exceeds $n = 52$.

(V3) External answer keys. Three exhibits printed in [9] were reproduced independently: their $n = 40$ maximal triple $\lambda = (9, 7, 6, 5, 4, 3, 2, 2, 1, 1)$, $\mu = (6, 4, 3, 2, 2, 1)$, $\nu = (6, 5, 4, 3, 2, 1, 1)$ has $c_{\mu\nu}^\lambda = 1484 = C(40)$ with ν/μ four unit squares; their counterexample at $n = 11$ to the nestedness conjecture of [8] has $c_{(4,1),(3,2,1)}^{(5,3,2,1)} = 3 = C(11)$ with $\mu \not\subseteq \nu$, and our brute force at $n = 11$ independently finds part 1 of Conjecture 1.1 failing there; and their four maximizers

at $n = 20$ all evaluate to $18 = C(20)$, which our brute force at $n = 20$ independently returns together with exactly four maximal triples. Finally, in the range $2 \leq n \leq 28$ covered by (V1) the uniqueness assertion of Conjecture 1.1 fails for the last time at $n = 27$ and holds at $n = 28$, which is where [9] place the threshold; that threshold was not an input to our computation.

(V4) A second Littlewood–Richardson implementation. The coefficient of the maximal triple at each of $n = 46, \dots, 53$ was recomputed by direct enumeration of Littlewood–Richardson skew tableaux, in a from-scratch counter sharing no code and no algorithm with `lrcalc` (script `crosscheck.py`). All eight agree, as do four textbook sanity values. In addition, `lrcalc` was used to recompute the coefficient of every one of the fourteen maximal triples found at $46 \leq n \leq 53$, not only the eight representatives printed below.

(V5) A second enumeration of the terminal vectors. Pass 2 was rerun from an independent script (`indep53.py`) which enumerates the terminal vectors directly, as the partitions of n with pairwise distinct odd parts, instead of by iterating T over all $p(n)$ sum vectors, and which shares no code with `maxlr.py`. It has now been run to completion at *every* degree $46 \leq n \leq 53$, and it reproduces Table 1 and Table 2 entry for entry: the value $C(n)$, the number of terminal classes, the number of classes attaining the maximum, and the largest value strictly below it all agree at all eight degrees. At $n = 46$, for instance, it returns $C(46) = 5932$ from the same 12 306 terminal vectors with one class attaining and 5 862 below, and at $n = 53$ it returns $C(53) = 31264$ from the same 32 490 terminal vectors with one class attaining and 27 046 below. The two enumerations were also verified to produce literally the same set of terminal vectors at every $46 \leq n \leq 53$.

What none of these gates provides is a second, independent confirmation of the *upper* bound at the new degrees from a different tool chain: the statement that no pair at degree n beats the reported value rests, for $46 \leq n \leq 53$, on Proposition 2.2 together with runs of pass 2 over the terminal vectors, by two scripts but with the same underlying Littlewood–Richardson library. The lower bounds, that the reported triples achieve the reported coefficients, are confirmed by two independent implementations.

4 Results

n	$C(n)$	terminal vectors	suspect pairs	maximal triples	orbits	runtime
46	5 932	12 306	8 477	1	1	354 s
47	8 481	14 193	12 032	2	1	590 s
48	9 926	16 357	14 304	2	1	849 s
49	12 828	18 803	14 884	2	1	1 345 s
50	15 762	21 581	23 084	2	1	2 239 s
51	18 570	24 766	8 910	2	1	11 209 s
52	22 982	28 394	4 901	2	1	9 922 s
53	31 264	32 490	2 835	1	1	13 429 s

Table 1: The eight new values. “Maximal triples” counts every triple attaining $C(n)$ with $|\mu| \leq |\nu|$; “orbits” counts them up to the interchange and conjugation action of Conjecture 1.1. Runtimes are a single core of a consumer laptop (Apple M4, Section 5) which was carrying unrelated load throughout, which is why $n = 51$ exceeds $n = 52$.

Representatives of the unique maximal orbit at each degree:

$$\begin{aligned}
n = 46 : \quad & \lambda = (10, 8, 7, 6, 4, 4, 3, 2, 1, 1), \quad \mu = \nu = (7, 5, 4, 3, 2, 1, 1); \\
n = 47 : \quad & \lambda = (10, 8, 7, 6, 5, 4, 3, 2, 1, 1), \quad \mu = (7, 5, 4, 3, 2, 1, 1), \quad \nu = (7, 6, 4, 3, 2, 1, 1); \\
n = 48 : \quad & \lambda = (10, 9, 7, 6, 5, 4, 3, 2, 1, 1), \quad \mu = \nu = (7, 6, 4, 3, 2, 1, 1); \\
n = 49 : \quad & \lambda = (11, 9, 7, 6, 5, 4, 3, 2, 1, 1), \quad \mu = (7, 6, 4, 3, 2, 1, 1), \quad \nu = (8, 6, 4, 3, 2, 1, 1); \\
n = 50 : \quad & \lambda = (12, 9, 7, 6, 5, 4, 3, 2, 1, 1), \quad \mu = \nu = (8, 6, 4, 3, 2, 1, 1); \\
n = 51 : \quad & \lambda = (11, 9, 7, 6, 5, 4, 3, 2, 2, 1, 1), \quad \mu = (7, 5, 4, 3, 2, 1, 1), \quad \nu = (8, 6, 5, 3, 2, 2, 1, 1); \\
n = 52 : \quad & \lambda = (11, 9, 8, 6, 5, 4, 3, 2, 2, 1, 1), \quad \mu = (7, 5, 4, 3, 2, 1, 1), \quad \nu = (8, 6, 5, 4, 3, 2, 1); \\
n = 53 : \quad & \lambda = (11, 9, 8, 6, 5, 4, 3, 3, 2, 1, 1), \quad \mu = (7, 5, 4, 3, 2, 1, 1), \quad \nu = (8, 6, 5, 4, 3, 2, 1, 1).
\end{aligned}$$

In every case ν/μ is a disjoint union of unit squares no two of which share an edge (at $n = 46, 48, 50$ it is empty, $\mu = \nu$, vacuously such a union), confirming part one of Conjecture 1.1; the orbit count 1 in Table 1 is part two. The second maximal triple at the six degrees where there are two is the conjugate of the first, and is printed in the run logs. The largest consecutive ratio in the extended table is $C(47)/C(46) \approx 1.430$ and the smallest is $C(48)/C(47) \approx 1.170$, a spread comfortably inside the range 1.037 to 1.576 of the consecutive ratios in $24 \leq n \leq 45$: the growth stays as irregular as it was.

The gap below the maximum

n	$C(n)$	terminal vectors attaining	largest value below	ratio
46	5 932	1	5 862	1.0119
47	8 481	2	7 154	1.1855
48	9 926	2	9 413	1.0545
49	12 828	2	12 449	1.0304
50	15 762	2	14 631	1.0773
51	18 570	2	18 356	1.0117
52	22 982	2	22 351	1.0282
53	31 264	1	27 046	1.1560

Table 2: The largest value of M over terminal vectors strictly below $C(n)$. By Lemma 2.5 the “attaining” column is a sum of T -cycle lengths; here it is one cycle throughout.

The compression framework makes one statistic cheap to record: the largest value $M(w^*)$ over terminal vectors *strictly below* $C(n)$. The ratio column of Table 2 is $C(n)$ divided by that runner-up. Every row comes from `top2_gap.py`, and the $n = 53$ row is additionally confirmed by the independent terminal enumeration of gate (V5), whose per-vector maxima give the runner-up directly; the two agree on all four quantities (logs `top2_gap_53.log` and `indep53.log`). The gap is uniformly positive but far from uniform in size: at $n = 46$ and $n = 51$ the runner-up comes within 1.2% of the maximum, while at $n = 47$ the winner exceeds the runner-up by 18.6%. Uniqueness up to symmetry holds at every computed n nonetheless, including the two tight degrees.

We stress what this statistic is not. Proposition 4.1 of [9] shows that if $C(n)$ is attained at a *single* k among the values $\{C(n, k) : 0 \leq k \leq n/2\}$, then part one of Conjecture 1.1 holds at n

(part one, not the uniqueness part), and [9] then speculate that a runaway growth of $C(n)$ might make that hypothesis provable asymptotically. The relevant gap there is between the largest and the second largest entry of the row $\{C(n, k)\}_k$, at most $n/2$ numbers indexed by k . That is a different quantity from the one in Table 2, and the present method does not deliver it: the compression map does not preserve $\min(|\mu|, |\nu|)$, so the terminal vectors do not respect k and no row of $C(n, k)$ can be read off them. (The step $w \mapsto w^*$ changes $\min(|\mu|, |\nu|)$ for 74% of the sum vectors at $n = 20$ and 82% at $n = 30$.) No row of $C(n, k)$ beyond $n = 45$ is computed here, and we are not aware of one anywhere.

Remark 4.1. The method is a search space reduction, not a new upper bound, and the cost stays superpolynomial. Already the number of partitions of n with distinct odd parts, which at every computed degree is exactly the number of terminal vectors, grows like $e^{\Theta(\sqrt{n})}$ just as $p(n)$ does, though on the computed range the terminal count sits between a tenth and an eighth of $p(n)$. Pass 2 parallelizes trivially over terminal vectors, and a multi-process or PyPy implementation would push the table a few terms further; we have not established how far.

5 Reproducibility

All computations were run with CPython 3.9.6 on macOS 15.7.1, on one core of an Apple M4 laptop (4 performance and 6 efficiency cores, 16 GB RAM), in a single process and a single thread, with the `lrcalc` library of Buch [5], version 2.1, through its Python bindings (`pip install lrcalc`). Wall-clock times were taken under unrelated load, so read them as upper bounds; the machine-independent measure of the work is the terminal-vector and suspect-pair counts in Table 1. The scripts and the raw logs of every run reported here are supplied as ancillary files with this submission, in the directory `anc/`:

file	what it does
<code>maxlr.py</code>	the three-pass algorithm; <code>python3 maxlr.py 46 ... 53</code>
<code>crosscheck.py</code>	independent LR counter, no <code>lrcalc</code> ; <code>python3 crosscheck.py</code>
<code>bruteforce_gate.py</code>	gate (V1); <code>python3 bruteforce_gate.py 2 28</code>
<code>indep53.py</code>	independent pass 2; <code>python3 indep53.py 53 3</code>
<code>top2_gap.py</code>	Table 2; <code>python3 top2_gap.py 46 ... 53</code>
<code>results_46_50.log</code>	runs behind Table 1, $n = 46$ to 50
<code>results_51_52.log</code>	runs behind Table 1, $n = 51, 52$
<code>results_53.log</code>	run behind Table 1, $n = 53$
<code>revalidate_24_45.log</code>	gate (V2), the reported range
<code>bruteforce_gate.log</code>	gate (V1)
<code>crosscheck.log</code>	gate (V4)
<code>indep_46_52.log, indep53.log</code>	gate (V5)
<code>top2_gap.log, top2_gap_53.log</code>	Table 2
<code>README.txt</code>	the above, with the environment and expected runtimes

Every value in the tables above appears in one of those logs, and the remaining figures quoted in the text are arithmetic on them. The range-based scripts take a degree range on the command line as shown above (`crosscheck.py` takes no arguments) and print one line per degree; none of them needs an argument file, a data file or a network connection.

Acknowledgements

Claude (Anthropic) was used for assistance with the implementation, the computations, and the preparation of this manuscript; every new computational result reported in this paper is an output of the accompanying code and its committed logs. The author reviewed the proofs, the code and the logs, and is responsible for the results and conclusions.

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